Sanshit Sharma | Senior GenAI Platform Engineer

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SUMMARY

GenAI software engineer with 12+ years building distributed systems at AWS, Cisco, and Qualcomm. Currently building Amazon Bedrock's core infrastructure; Model Migration, Model Evaluation and ML pipelines serving thousands of enterprise customers. Combines deep systems expertise with AI-assisted development practices to accelerate delivery velocity while maintaining production-grade quality. Shipped multiple AWS re:Invent launches (2023, 2025). Patent holder.

TECHNICAL SKILLS

Languages: Java, Python, GoLang

AWS & AI/ML: Bedrock, SageMaker, Lambda, Step Functions, CDK, CloudWatch, DynamoDB, S3

Infrastructure: Distributed Systems, State Machines, CI/CD Pipelines, Microservices

Data & Messaging: Kafka, gRPC, REST APIs, PostgreSQL

Development: AI-assisted development (Kiro, Claude Code, Wasabi), rapid prototyping, test-driven development

PROFESSIONAL EXPERIENCE

Software Development Engineer II | March 2022 - Present
Amazon Web Services (AWS) — San Francisco Bay Area

Bedrock Model Migration Service (re:Invent 2025)

- Designed generic prompt extraction framework using JSON schemas to parse any InvokeModel payload, eliminating hardcoded model-specific parsers. Reduced time to onboard new models from 4 weeks to 2 days.
- Owned end-to-end workflow orchestration—designed and implemented Step Functions & Lambda definitions and integrations across the 3-stage migration workflow
- Defined AppSec-approved IAM policy templates enabling one-click execution role creation. Required deep cross-service expertise to define least-privilege permissions that protect customer accounts while ensuring runtime reliability.

Model Migration Ops Tool

- Proactively designed and built a greenfield AI-powered Ops Tool, consolidating job telemetry across 3 service accounts from Step Functions, Lambdas, and CloudWatch into a unified debugging interface
- Implemented AI-driven root cause analysis: Users click "Analyze with AI" to send error logs to a model with a custom few-shot learning prompt, identifying primary failure causes and tertiary issues. Reduced MTTR by 80%

RAI Assessments Production Pipeline

- Transformed research package into production pipeline across 12 AWS regions. Led 2 L4 engineers building CI/CD with integration tests
- Achieved 77% container image reduction (3GB → 700MB) and 80% build time improvement (~10 min → <2 min)

RFT Evaluation Framework

- Designed and implemented an end-to-end evaluation framework for Model Customization's flywheel in 1.5 weeks. Leveraged AI-assisted development (Kiro) to rapidly prototype and iterate on CDK infrastructure, state machine definitions, and Lambda handlers

Bedrock Model Evaluation (re:Invent 2023)

- Designed and developed the control plane and APIs for model evaluations, enabling scalable evaluation processes for foundation models on Bedrock

Senior Software Engineer | May 2015 - March 2022
Cisco Systems — San Jose, California

- Technical lead for Crosswork Change Automation—designed cloud-deployable application for closed-loop automation in network data centers leveraging streaming telemetry for real-time device monitoring (Patent granted)
- Delivered REST microservice application for change automation supporting pre/post checks with automatic rollback

- Developed key modules for streaming telemetry in Cisco devices (C, C++, Python); presented at Cisco Live 2016
- Created Cython interface wrapping C++ DME libraries for Python telemetry; integrated with Splunk and Elasticsearch

Software Development Engineer | January 2013 - May 2015
Qualcomm Technologies Inc. — San Diego, California

- Software development in C/C++ for mobile device modems across cellular protocols (GSM, LTE, Voice Over LTE)
- Led software commercialization for customers in North America and Asia Pacific

PATENTS & PUBLICATIONS

Patent: *"Continuous Monitoring of Network Devices During Maintenance" — US Patent Application US20200358648A1, Cisco Technology Inc., Filed October 2019*

Publication: *"T-Hex: True Hexagonal Regular Topology Formation in Large Scale Wireless Sensor Networks" — IEEE MASS 2012*

EDUCATION

M.S., Computer Science — University of Cincinnati, Ohio (GPA: 3.82) — November 2012

B.Tech, Computer Science and Engineering — MANIT, Bhopal, India — May 2008

ACHIEVEMENTS

- Multiple re:Invent launches on Amazon Bedrock (2023, 2025) — AWS's flagship GenAI platform
- Certified Amazon Security Guardian: Helped multiple teams build secure architecture. Engagement from design to delivery.
- Five Cisco Connected Recognition awards for project contributions
- Two Qualcomm Star Awards for GSM/TDS modem development and SGLTE commercialization